Automation Framework Case Study: SauceDemo

Software Testing Lab (R5CO4101L) · Selenium automation + basic performance testing · November 2026 · github.com/Abhinavmadake/saucedemo-framework

Team: Abhinav (231070028), Amit (231070020), Chaitanya (231070066), Prasanna (231070024), Ritesh (231070053), Yogesh (231070037)

Scope

AUT: `https://www.saucedemo.com`, user `standard_user`. Nine functional requirements automated: the six mandatory ones (FR-01, 02, 07, 12, 14, 16) and three extra (FR-04, 09, 11; FR-04 is the extra negative case). Three performance requirements (NFR-01 to 03) tested with JMeter. Test cases derived with Equivalence Partitioning and Boundary Value Analysis.

Framework

Python 3.11+, Selenium 4.52, PyTest 9.2, pytest-html 4.2, webdriver-manager 4.1, Apache JMeter 5.6.3.

- **Page Object Model:** locators and actions live in `pages/`; tests only call page methods.
- **Explicit waits only** (`WebDriverWait`), no `time.sleep`; waits retry when the page re-renders an element.
- **Data-driven:** login and postal code cases read from CSV files in `testdata/`.
- **Markers:** `smoke` for happy paths, `regression` for the full suite.
- **Evidence:** screenshot on failure (saved and embedded in the HTML report), step log per test, HTML report.
- **Traceability:** every case has a TC ID mapped to its FR and test function in `docs/traceability.csv`.

Structure

```
pages/        page objects (7 pages + base page)
tests/        one test file per module
testdata/     CSV test data
utils/        config, CSV loader, step logger
docs/         test cases, defects, traceability
performance/  JMeter plan, results, dashboard
reports/      HTML report, screenshots, logs
```

How to run

```
pip install -r requirements.txt
pytest -m smoke     smoke suite
pytest              full suite + HTML report

jmeter -n                          (one command)
  -t performance/jmx/saucedemo_weekend_lab.jmx
  -l performance/results/results.jtl
  -e -o performance/html_report
```

Functional results

Run 3 Nov 2026, 19 cases in under a minute; 5 smoke tests (`pytest -m smoke`).

FR	Requirement	Cases	Pass	Fail
FR-01	Login success	1	1	0
FR-02	Invalid password (CSV, 4 EP/BVA cases)	4	4	0
FR-04	Blank password	1	1	0
FR-07	Add to cart (cart size 0→1, 1→2, all 6)	3	3	0
FR-09	Cart matches Products (2 and 6 items)	2	2	0
FR-11	Navigate to checkout	1	1	0
FR-12	First Name required	1	1	0
FR-14	Postal code: 4/5/6 digits, non-numeric (CSV)	5	2	3
FR-16	Finish order, cart emptied	1	1	0
Total		**19**	**16**	**3**

The 3 failures are the FR-14 defects below; each has a screenshot in `reports/screenshots/` and in the HTML report.

Performance (JMeter)

20 users, 60 s ramp-up, 3 min, 0-500 ms think time. 10,434 requests at 58 req/s.

NFR	Metric	Threshold	Measured	Result
NFR-01	Login Page P95	< 3.0 s	0.044 s	**PASS**
NFR-02	Products Page P95	< 4.0 s	0.044 s	**PASS**
NFR-03	Error rate	< 2 %	0.60 %	**PASS**

All 63 errors were dropped connections; the site never returned a non-200 status. SauceDemo is a static app on GitHub Pages and login runs in the browser, so these numbers measure page delivery, not a backend.

Defects

ID	Summary	Severity
DEF-14-01	4-digit postal code (1234) is accepted	Medium
DEF-14-02	Alphabetic postal code (abcde) is accepted	Medium
DEF-14-03	Mixed postal code (12ab5) is accepted	Medium

The site only checks that Postal Code is not empty; FR-14 requires a numeric code of 5 or 6 digits.